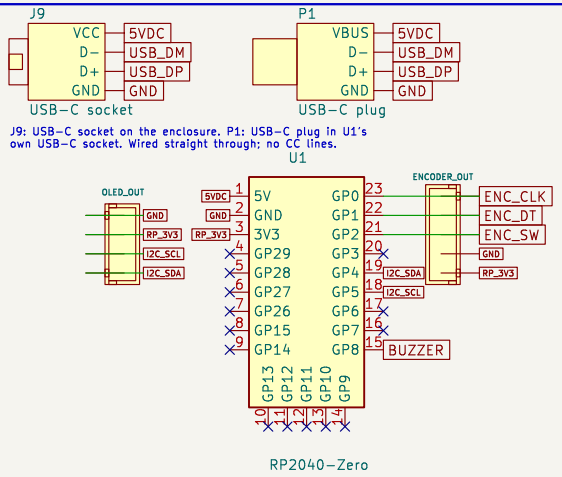
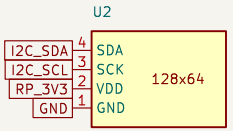
GERARD

RP2040-ZERO MICROCONTROLLER



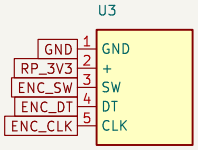
U1 is powered through its own USB-C socket, by P1.
5VDC is its 5V pad, joined to the USB-C supply.
Pin numbers count the edge pads from 5V round the board; the board prints only the GP names.

OLED DISPLAY



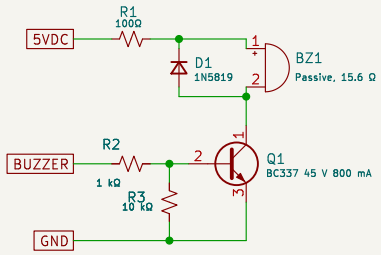
SSD1306, 128 × 64, white.

ROTARY ENCODER

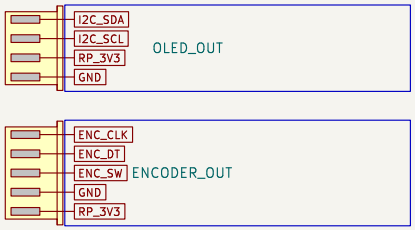


The module's own resistors pull CLK, DT and SW up to +.

MAGNETIC BUZZER AND DRIVER



Q1: find C, B, E with a tester before soldering; BC337 makers differ.
R1: 100 Ω or more at 0.25 W, so that it survives BUZZER stuck high with Q1 on.
D1: band (cathode) on BZ1's + lead.



NOTES

1. Power: only through the enclosure's USB-C socket (J9), 5 V DC.
2. Parts are through-hole on one perfboard: U1, BZ1 and its driver. U2 and U3 are cabled to it.

CABLES

JST XH at U1's board only: headers J1 and J5 there, housings J2 and J6 on the cables. At U2 and U3 the cables are soldered directly. Each end is drawn as built. Which pin of each connector is pin 1 is not recorded.
USB-C extension J9 to P1: 5VDC, GND, USB_DP, USB_DM.

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Title: GERARD

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